Daniel Felipe Forero Sánchez

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RESEARCH EXPERIENCE

Postdoctoral Researcher in Cosmology · 2024 – Present — *ICCUB, Universitat de Barcelona*

- Leading full-shape & higher-order-statistics cosmology with DESI: lead author of the joint DR1 Full-Shape + DR2 BAO constraints and of DESI's first joint bispectrum + power-spectrum full-shape analysis; led the validation of the covariance matrices adopted for the official DESI 2024 results (incl. fitting the DR1 EZmock ensembles).
- Lead of the **Holi mock suite** — full-sky galaxy lightcones for the DESI DR2/DR3 covariances (fast approximate gravity + differentiable, physically motivated cosmic-web bias). Contributor to **desilike** (ShapeFit bispectrum fitting; flow-based non-Gaussian posterior interpretation).
- Developing the arc from optimized compression (ShapeFit) toward learned compression and likelihood-free, field-level inference.

PhD Researcher in Cosmology · 2020 – 2024 — *EPFL, Laboratory of Astrophysics* (ad. J.-P. Kneib)

- Developed **CosmoMIA**, a physically motivated fast model of redshift-space structure (JCAP 2024); established **void-BAO** sensitivity to selection effects (MNRAS 2022) and ML corrections of **mass-resolution** systematics (MNRAS 2022); contributed to the DESI Mock Challenge and density-split clustering (SUNBIRD).

EDUCATION

PhD Physics — EPFL, Switzerland · 2020 – 2024 · Thesis: *Cosmology with the Large Scale Structure from Spectroscopic Surveys* (ad. J.-P. Kneib)
MSc Physics — EPFL, Switzerland · 2018 – 2020 · Thesis: *Robustness of void BAO to survey incompleteness effects*
BSc Physics + BSc Geoscience (double degree) — Universidad de Los Andes, Colombia · 2014 – 2018

SELECTED PUBLICATIONS *(35 total, 6 first-author — full list:* `publication_list.md` */ ORCID)*

- **D. Forero-Sánchez, H. Gil-Marín, L. Verde**, et al. (64 authors), "Cosmological constraints from a joint DESI DR1 Full-Shape and DR2 BAO", *Journal of Cosmology and Astroparticle Physics*, 2026. doi:10.1088/1475-7516/2026/06/043
- **Forero-Sánchez, D.**, Novell-Masot, S., Gil-Marín, H., "Cosmological constraints from the DESI DR1 Bispectrum Full-Shape and DR2 BAO", *arXiv e-prints*, 2026. arXiv:2606.23936
- **D. Forero-Sánchez, M. Rashkovetskyi, O. Alves**, et al. (47 authors), "Analytical and EZmock covariance validation for the DESI 2024 results", *Journal of Cosmology and Astroparticle Physics*, 2025. doi:10.1088/1475-7516/2025/04/055
- **D. Forero Sánchez**, F.-S. Kitaura, F. Sinigaglia, J.M. Coloma-Nadal, J.-P. Kneib, "CosmoMIA: cosmic web-based redshift space halo distribution", *Journal of Cosmology and Astroparticle Physics*, vol. 2024, no. 7, 2024. doi:10.1088/1475-7516/2024/07/001

- **Forero-Sánchez, Daniel**, Zhao, Cheng, Tao, Charling, Chuang, Chia-Hsun, Kitaura, Francisco-Shu, Variu, Andrei, Tamone, Amélie, Kneib, Jean-Paul, "Cosmic void baryon acoustic oscillation measurement: evaluation of sensitivity to selection effects", *Monthly Notices of the Royal Astronomical Society*, vol. 513, no. 4, pp. 5407–5422, 2022. doi:10.1093/mnras/stac1268
- **Daniel Forero-Sánchez**, Chia-Hsun Chuang, Sergio Rodríguez-Torres, Gustavo Yepes, Stefan Gottlöber, Cheng Zhao, "A machine learning approach to correct for mass resolution effects in simulated halo clustering statistics", *Monthly Notices of the Royal Astronomical Society*, vol. 513, no. 3, pp. 4318–4331, 2022. doi:10.1093/mnras/stac1239
- **In prep. (lead):** *The Holi mock suite — full-sky galaxy lightcones to $z\approx4$ for DESI* (JCAP).
- Co-author on $\sim$28 more, incl. the DESI DR1/2024 series, density-split clustering + SUNBIRD, 21-cm foreground removal with deep learning (MNRAS 2024), and the MUST & WST white papers.

SOFTWARE

Holi mocks (DESI DR2/DR3 full-sky lightcones) · **desilike** contributor (`cosmodesi`) · **CosmoMIA** · **pydive** (Delaunay void finder) · **geofptax** (JAX bispectra) · **pyfcfc** (fast 2PCF) — github.com/dforero0896

RESEARCH INTERESTS

Cosmology – Large-Scale Structure – Spectroscopic Surveys – Fast Simulations & Mock Generation – Covariance & Full-Shape Statistics – Statistical / Machine Learning for Cosmology – Voids & Alternative Statistics

SELECTED TALKS

"Projection-free cosmology constraints with DESI DR1 full-shape and DR2 BAO" — XI Meeting on Fundamental Cosmology (Santander, 2025); EAS Meeting (Lausanne, 2026). · *"Holi mock suite: 1000 lightcones for DESI DR2/3"* — DESI Summer Collaboration Meeting (Durham, 2026). · *"Delaunay-triangulation spheres as complementary tracers"* — COSMO23 (Madrid); DESI Summer (Durham); EAS 2023 (Kraków); WST (Vienna), 2023. · *"A new generation of fast mocks for covariance matrices"* — DESI Summer/Winter Collaboration Meetings (2022–23). · *"Improving DM-halo resolution for LSS using ML"* and *"Selection effects on void BAO"* — EAS 2022 (Valencia); Tenerife; Cosmology in Colombia (Bogotá), 2020–22.

TEACHING, MENTORING & SERVICE

Co-supervised 3 EPFL student projects (DL power-spectrum denoising; NN void inference; redshift errors & Σm_ν). Doctoral Assistant, EPFL (*Astrophysics IV*; *General Physics*, 2021–23); TA, Universidad de los Andes (2015–18). DESI working-group service (full-shape, covariances, mocks). **Co-lead, MUST Simulations Working Group**. Outreach: Geneva Observatory tours & talks (2021–24).

SKILLS

Python · Julia · C/C++ · Bash · **JAX** / **PyTorch** / **TensorFlow** (differentiable & flow-based modelling) · GPU & HPC/Slurm · perturbation theory & full-shape modelling · approximate simulations · statistical analysis & visualization · Git

LANGUAGES & REFERENCES

Spanish (native) · English (fluent) · French (fluent). — References on request: L. Verde & H. Gil-Marín (ICCUB), J.-P. Kneib (EPFL), F.-S. Kitaura (IAC).